

Instruction

Nichrome is an alloy of nickel and chromium. It may also be mixed with small amounts of other elements. Nichrome is used in a wide range of applications where high thermal resistance is required, such as electric heating wires, due to their low cost, high strength, high ductility, good oxidation resistance, and high melting point [1].

Historical development, chemical composition, and methods of fabrication

The earliest recorded origin of nichrome alloys was in 1906 when the American Albert Marsh applied for the earliest patent for a nichrome alloy. Then, nichrome electrothermal alloys were developed in the UK in the late 1930s. During World War II, nickel-chromium alloys were used extensively in aircraft turbine engines and, because of their excellent performance, were later used in rocket engines.

In general, the mass percentage of nickel in nichrome is 60%~80%, and the mass percentage of chromium is 10%~27%. Nichrome is mainly austenitic and martensitic in the microstructure. As the difference in electronegativity between nickel and chromium is only 0.2, the nichrome contains only metallic bonds [2].



Figure 1. Microstructure of nichrome [3]

There are many different methods of casting nichrome, but the general process is the same. All of them are made by first batching all the alloy components, then melting and smelting them and refining them twice. It is kept warm for a period before being forged.

Advantages and disadvantages of Nichrome alloys as materials

Advantages: Low manufacturing costs (suitable for all scenarios in life), high mechanical and creep strength, high ductility (easy to draw), corrosion resistance, oxidation resistance, high melting point, high-temperature stability (melting point of about 1,400 °C) Nichrome alloys are known for their high mechanical and creep strength.

Disadvantages: In terms of performance, nichrome is difficult to assemble in use and can cause short circuits. In terms of health and environment, firstly, the inhalation of nickel dust during the processing of nickel-chromium alloys can cause respiratory organ dysfunction and alveolar hypertrophy, which can be harmful to health. Secondly, current technology does not allow for the rapid and complete degradation of heavy metals, which can contaminate the earth and water, thus damaging the ecological environment.

Due to the high resistance of the nichrome, heat is generated even at low currents. It is therefore widely used in areas where heating is required, for example in ceramic production and glass production, where all components are connected using Inconel [4]. It is also commonly used as a central electrode in automotive spark plugs because of its extremely low resistivity ($112 \mu\Omega\text{-cm}$).



Figure 2. Nichrome spark plug [5]

Summary

Nichrome alloys were first introduced in the First World War and have undergone continuous exploration and refinement, resulting in an increasing range of applications and functional properties. Nichrome is mostly austenitic, thermally corrosive and is often used as thermistors. Powder metallurgy is the main method of producing nichrome. In addition, the degradation of heavy metals, which are difficult to degrade and are harmful to humans and the environment, needs to be improved.

References

1. Jounoy B. What is Nichrome Wire used for? [Internet]. Sciencing. 2020 [cited 2023Feb28]. Available from: <https://sciencing.com/nichrome-wire-used-for-5871336.html>
2. Flournoy B. What is Nichrome Wire used for? [Internet]. Sciencing. 2020 [cited 2023Feb27]. Available from: <https://sciencing.com/nichrome-wire-used-for-5871336.html>
3. Microstructures - surface coatings and heat resisting alloys - lesson-3 - specimen03. [cited 2023Feb27]. Available from: <https://www.georgesbasement.com/Microstructures/CastIronsHighAlloySteelsSuperalloys/Lesson-3/Specimen03.htm>
4. Resistance wire: High resistivity wire: Specialty wire: MWS Wire [Internet]. MWS Wire - Magnet Wire, Specialty Wire. 2022 [cited 2023Feb28]. Available from: <https://mwswire.com/resistance-wire-2/>
5. GmbH DN. Search results [Internet]. Spark Plug Alloys | Deutsche Nickel GmbH, Germany. [cited 2023Feb28]. Available from: <https://www.deutsche-nickel.de/en/products/wire/spark-plug-alloys>